

CitrusAd: The world's fastest growing retail media platform running on Google Cloud

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CitrusAd transforms retail media landscapes for retailers and suppliers, running its platform on Google Cloud and leveraging BigQuery Cloud SQL for serverless analytics.



About CitrusAd

CitrusAd is being recognized as the world's leading white-label self-serve retail media platform that enables retailers to monetize their digital shelf space while enabling suppliers to increase sales by launching cost-effective digital campaigns right at the point of purchase. Since launching in 2017, CitrusAd has become the global retail industry's preferred sponsored product and banner ad platform. Successful retailers across all verticals from 25 countries are leveraging the CitrusAd platform, serving 50 billion ad requests annually to create more

Google Cloud results

- Serves ads to shoppers in milliseconds with Google Cloud Platform for a frictionless shopping experience
- Reduces release cadence from six weeks to fortnightly for faster feature deployment to customers
- Easy and independent microservices deployment speeds up campaign time to market

Serve
hundreds
of thousands
of ads
simultaneously
to see
powerful
media
leading
around the
world

personal shopping experiences and deliver greater ROI for brands.

Location: Australia

Kubernetes Engine

Google Cloud (<https://cloud.google.com/>) Kubernetes Engine),

Google Cloud console

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Google Cloud console

(<https://cloud.google.com/cloud-console/>)

Retailers worldwide have experienced a dramatic shift from store to ecommerce sales as consumers spend more time at home because of COVID-19. According to Bain & Company, 45% of US consumers

(<https://www.bain.com/insights/one-time-offer-grocers-have-a-unique-opportunity-to-reset-promotions/>)

are now buying groceries online instead of shopping in store as they did before the pandemic. In the US, the National Retail Federation noted that online and non-store sales

(<https://nrf.com/media-center/press-releases/april-retail-sales-drop-nearly-twice-much-march-during-coronavirus>)

during April 2020 increased 21% year on year. As a result, brands are prioritizing digital marketing such as paid search instead of in-store promotions to drive sales and awareness.

To stay competitive, retailers need a robust ad solution to attract online revenue streams while ensuring that they deliver a personalized online shopping experience. Enter CitrusAd

(<https://www.citrusad.com/>), a retail media platform that helps retailers seamlessly monetize their digital page views by serving relevant sponsored products and banner ads to shoppers as they enter key search terms and browse category pages online to enhance their shopping experience.

Founded in 2017, CitrusAd's network grew from 968 to 23,350

(<https://www.businesswire.com/news/home/20200727005539/en/>)

retail stores in just three months during COVID-19 lockdown. The company turned to Google Cloud

(<https://cloud.google.com/>) to ensure it could scale quickly without incurring too much cost. CitrusAd uses Google Kubernetes Engine (/kubernetes-engine) (GKE) to serve millions of ads daily to online shoppers and BigQuery (/bigquery) and Cloud SQL (/sql) for retail analytics.

“Compared to our legacy cloud service provider, Google Cloud is easier to operate and maintain,” says Adam Skinner, Chief Technology Officer at CitrusAd. He adds that it also offers more automation tools such as automated deployment for Google Kubernetes Engine. By streamlining its cloud operations, the company can focus on building new features such as a self-service architecture that will provide retailers with more flexibility and control.

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—Adam Skinner, Chief
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CitrusAd

Speeding up customer engagement with microservices

A key proposition for CitrusAd is its ability to simplify and streamline the online monetization of a retailer’s digital assets so they can focus on their core business. Using the CitrusAd API, retailers can start to sell advertising on their ecommerce website, without investing in an in-house solution and workforce. The simple self-managed platform empowers brands and agencies to create and manage their campaigns and budgets in minutes.

The system architecture is a collection of microservices that sit on a cluster in Google Kubernetes Engine. These microservices power functions within CitrusAd such as ad serving, ad management, campaign management, catalog management, attribution, and reporting.

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With the understanding that online shoppers may go to a competitor’s website if they don’t find what they want quickly, CitrusAd aims to help retailers close sales by serving relevant ads on top of the category and search results. This allows consumers to enjoy a seamless shopping experience because the personalized ads and organic results load at the same time. “The flexibility of microservices allows us to customize the relevancy algorithm to meet the specific needs of each retailer,” says Adam. “It could take months for a retailer to create and implement a similar algorithm on their own.”

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Real-time transparent reporting for retail success

Like eye-level shelf space in the retail store, brands at the top of search results get the most awareness and conversions on a retail website by attracting the customer’s attention. CitrusAd helps retailers and advertisers optimize their campaigns with data insights on how ads are performing on this digital shelf space.

On the CitrusAd reporting dashboard, retailers can drill down on revenue by campaigns, products, keywords, and teams. Advertisers can track what search terms shoppers are using to find their products, how they are ranked on the results pages, and how their ad performed against competitors.

CitrusAd uses its existing business intelligence solution to integrate historical data from Cloud SQL

and big data from BigQuery, turning them into live visualizations. CitrusAd's media sales team, retailers, and suppliers also use the CitrusAd reporting dashboard to support business decisions.

"The combination of BigQuery and our existing business intelligence solution is like a match made in heaven. We can perform queries with BigQuery, and because it integrates so easily with our other tools, we can pull the results for visualization and let users view data at any time," says Adam.

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Simplifying cloud management with DevOps tools

CitrusAd’s tech team used a lift-and-shift approach to seamlessly migrate all its workloads to Google Cloud, without downtime. The company currently operates in four Google Cloud zones, specifically in Australia, Asia, Europe, and the United States.

“We offer peace of mind to retailers by operating each zone on Google Cloud independently to comply with specific data requirements such as GDPR and CCPA,” says Adam. “At the same time, the Kubernetes architecture gives us the ability to share global resources between zones.”

CitrusAd's team finds that cloud management is much more intuitive on Google Cloud. From the [Google Cloud console](/cloud-console) (/cloud-console) web admin user interface, CitrusAd can quickly deploy and manage all cloud resources, from data analysis in BigQuery to developer services in GKE. The ease of management and visibility effectively aligns with the company's strategic needs.

"We're setting up CI/CD pipelines to automate deployments on Google Kubernetes Engine. Automated features such as one-click deployment reduce our release cadence from once every six weeks to daily," says Adam. "That gives us the ability to roll out new features to retailers much faster and maintain a stable workload within Google Cloud."

Driving the future of retail transformation with machine learning

CitrusAd is exploring a machine learning model that feeds back into BigQuery for better prediction of sales. The model looks at historical data and real-time data to predict campaign success. Retailers can see the return on advertising spend (ROAS) prediction for one search term versus another and make an informed decision based on data.

"Our customers want to work with the most innovative service providers out there, so we need to stay two steps ahead for our partners and prospects at all times," says Adam. "The forward-looking approach of Google Cloud, together with Kubernetes

and containerization, will help us keep innovating so we can serve our customers better every day.”